COAL

Graded Lab - 2

By

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CODE

```
.model small
.stack 100h
;Write an assembly program that declares and initializes an array of 10
numbers.
;The program should count all even numbers in the array and store the
result in
;a variable named EVEN_COUNT.

.data
    ;an array of 10 elements
    array db 0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08, 0x09, 0x0a
    ;a variable to store total count of even number occurances
    EVEN_COUNT dw 0x0000
.code
    main proc
     mov ax, @data
     mov ds, ax ;initilize the data segment

     mov dl, 0x02 ;mov 2 into dl to divide
     mov cx, 10 ;loop counter
     mov si, 0 ;array index pointer
     mov bx, 0x0000 ;to store total number of occurances of even numbers

     evenCount:
         mov al, [array + si]
         mov ah, 0
```

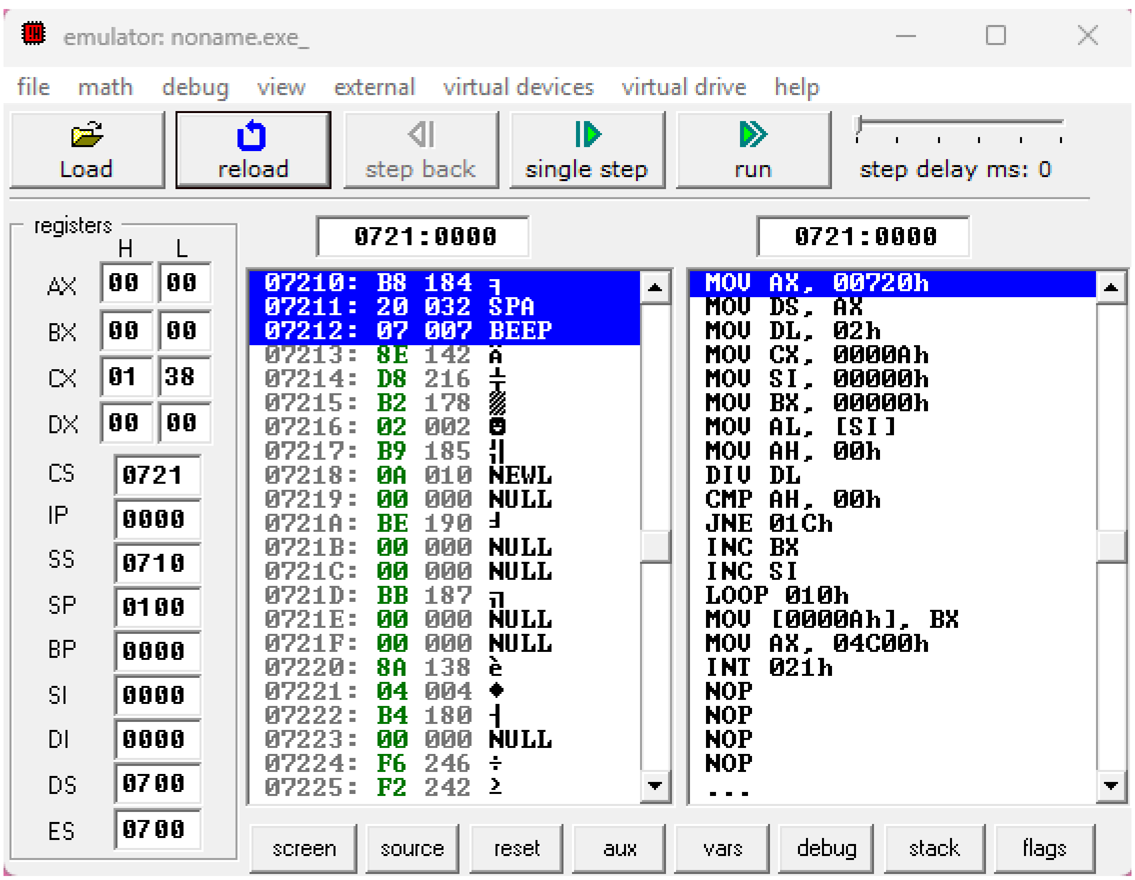
```
        div dl ;divede by 2

        cmp ah, 0 ;compare to find if division by 2 gives any remainder
        jnz notCount ;if odd jmp

        inc bx

    ;label which will run if the pointer is on an odd number
    notCount:
        inc si

    loop evenCount

    ;load bx(total evens in array) into variable
    mov EVEN_COUNT, bx

   main endp
.exit
```

ScreenShots

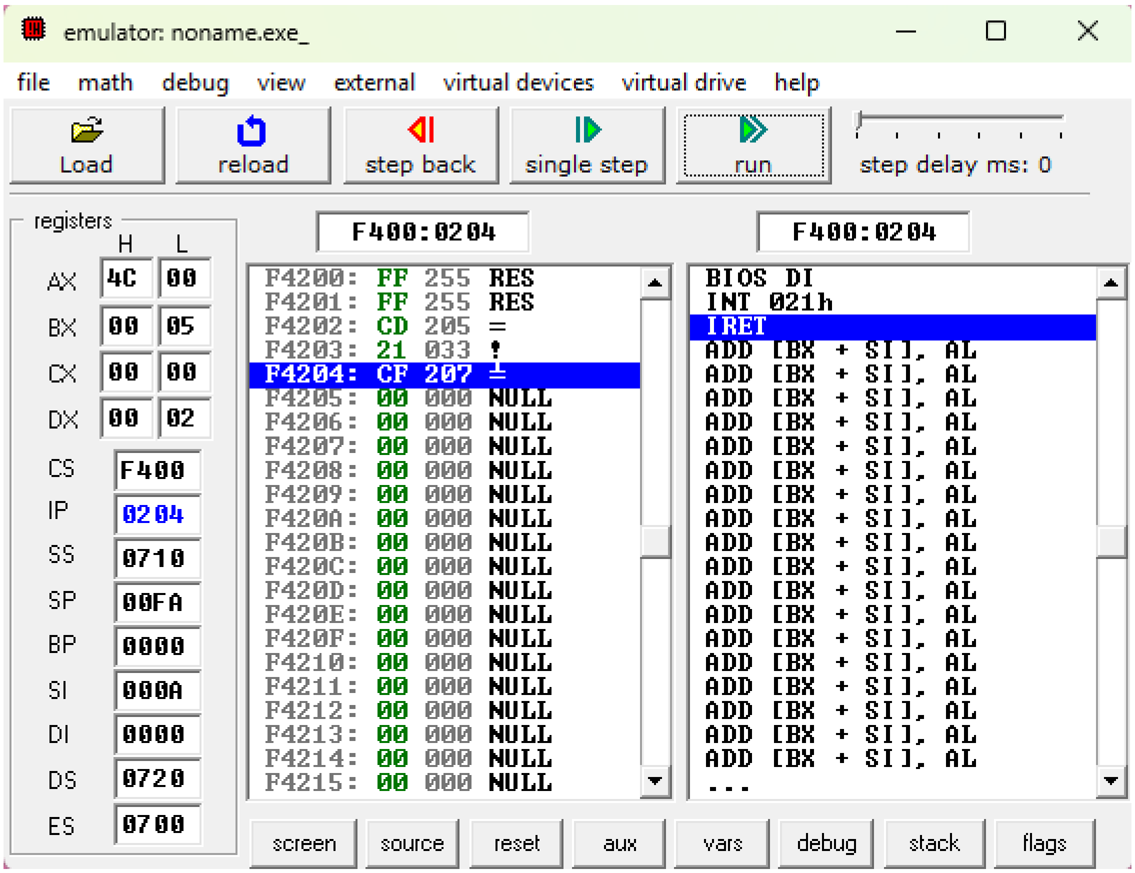
Initilized array and variable in data segment

```
0720:0000   01 02 03 04 05 06 07 08-09 0A 00 00 00 00 00 00
0720:0010   B8 20 07 8E D8 B2 02 B9-0A 00 BE 00 00 BB 00 00
0720:0020   8A 04 B4 00 F6 F2 80 FC-00 75 01 43 46 E2 F1 89
0720:0030   1E 0A 00 B8 00 4C CD 21-90 90 90 90 90 90 90 90
0720:0040   90 90 90 90 90 90 90 90-90 90 90 90 F4 00 00 00
0720:0050   00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0720:0060   00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0720:0070   00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
```

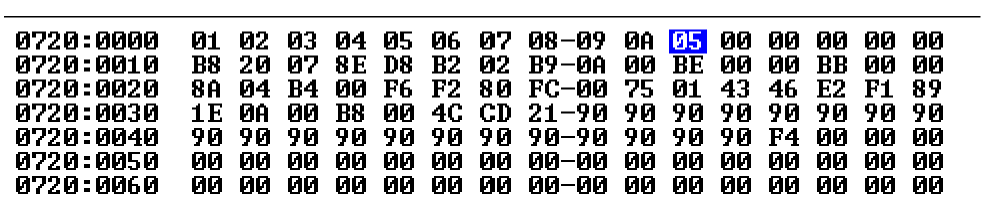
Values at start



Values at end



Updated memory with EVEN_COUNT showing total even numbers found in array



```
0720:0000  01 02 03 04 05 06 07 08-09 0A 05 00 00 00 00 00
0720:0010  B8 20 07 8E D8 B2 02 B9-0A 00 BE 00 00 BB 00 00
0720:0020  8A 04 B4 00 F6 F2 80 FC-00 75 01 43 46 E2 F1 89
0720:0030  1E 0A 00 B8 00 4C CD 21-90 90 90 90 90 90 90 90
0720:0040  90 90 90 90 90 90 90 90-90 90 90 90 F4 00 00 00
0720:0050  00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0720:0060  00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
```